

Surface-Enriched Amphiphilic Polysiloxane Coating with Superior Antifouling Ability and Strong Substrate Adhesion

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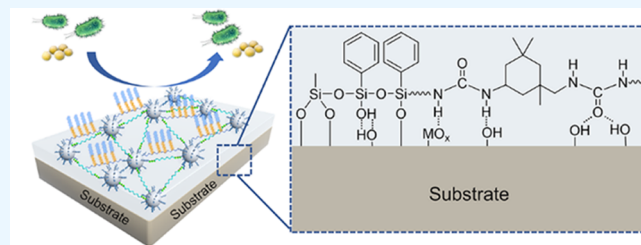
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ABSTRACT: The coexistence of superior antifouling ability and strong substrate adhesion is a typical trade-off for a silicone coating but is highly desirable in numerous applications. Herein, we present a surface-enriched antifouling coating consisting of a bis-silane-terminated polyurea (SPU), an oligosiloxane nanocluster, and a bi-silanol-terminated poly(dimethylsiloxane) with fluorocarbon and poly(ethylene glycol) side chains as a reactive amphiphilic polymer (RAP). The oligosiloxane nanoclusters can cross-link the flexible SPU into a tough polymer network and interlink the polymer network to the substrate, whereas the low-surface-energy RAP can self-enrich on the surface during coating formation. Such a polysiloxane coating exhibits high transparency (>85% transmittance), high tensile strength (~12 MPa), superior fouling resistance against proteins and bacteria, and strong adhesion strength on various substrates including glass, ceramic, steel, Ti, and epoxy (3–15 MPa). This research provides a universal and tunable approach for the future design of antifouling coatings in flexible electronics, medical devices, and marine industries.

KEYWORDS: antifouling coating, polysiloxane, oligosiloxane nanocluster, self-enriched, strong adhesion



✓ Transparent ✓ Strong yet tough ✓ Superior antifouling ✓ Strong adhesion

INTRODUCTION

Microbial adhesion onto surfaces is detrimental to flexible electronics, biomedical engineering, the maritime industry, and other applications.^{1–4} Thus, the design and preparation of antifouling coatings have received immense attention in both academia and industry.⁵ The most common method of preventing biofouling is to use toxic metals or organic compounds in coatings, which always impact untargeted objects.^{6,7} Thus, considerable effort has been made to develop eco-friendly yet efficient antifouling coatings, such as silicone-based fouling release coatings,⁸ amphiphilic copolymers,⁹ protein resistance polymers,¹⁰ degradable polymers,¹¹ and biomimetic polymers.¹² Among them, biocide-free poly(dimethylsiloxane) (PDMS)-based coatings have received great attention and are applied to fouling release and drag-reducing of the ship hull surface due to their low surface energy and low modulus.¹³ Nevertheless, most silicone coatings are based on linear polysiloxanes with high chain flexibility and low surface energy; thus, it is mutually exclusive to achieve the coexistence of superior antifouling ability and strong substrate adhesion.^{14,15}

Using different groups (epoxy, catechol, urethane, urea, 2-ureido-4[1H]-pyrimidine, etc.)^{16–19} to modify the linear PDMS elastomer can improve their substrate adhesion and mechanical properties, but they usually sacrifice antifouling ability due to surface changes caused by those polar groups. Adding inorganic particles (SiO₂, nanodiamond, graphene

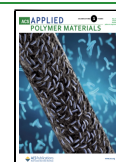
oxide, multiwalled carbon nanotubes, etc.) to improve substrate adhesion often increases surface roughness, thus weakening the antifouling performance.^{20–23} On the other hand, the introduction of fouling-resistant groups [poly(ethylene glycol) (PEG), zwitterions, etc.] can effectively prevent biofouling by forming a hydration layer on the surface.^{24–27} However, these additives usually have poor compatibility with the PDMS elastomer due to their vastly different solubility parameters and surface energy. Moreover, they are generally incorporated into coatings in a large proportion to exert antifouling performance, which is easily trapped in bulk, thus weakening the mechanical properties and substrate adhesion.^{28–32} Thus, simultaneously achieving excellent antifouling, substrate adhesion, and mechanical properties is a significant challenge for a silicone coating.

Silane-terminated polyurethane/polyether/polyurea has been widely used as adhesives or sealants in the construction, automotive, and electronic industries due to its strong substrate adhesion, elasticity, water resistance, and moisture-curable characteristics.^{33–36} Moreover, these polymers con-

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taining silane groups permit the formation of reactive silanol groups that can be used to cross-link other materials or form chemical bonding with substrates.^{37–40} Therefore, introducing these silane-terminated polymers into a polymer network may be feasible to improve substrate adhesion but not decline other properties. Herein, we present a polysiloxane coating with antifouling ability and strong substrate adhesion by combining bis-silane-terminated polyurea (SPU) with stereoscopic oligosiloxane nanoclusters and a small amount of antifouling reactive amphiphilic polymer (RAP) *via* sol–gel chemistry. The bis-silane-terminated polyurea containing silane groups permit the formation of reactive silanol groups that can be used to connect other materials, while urea groups can form strong hydrogen bonding with other polymer chains or substrates. The oligosiloxane nanoclusters are synthesized by partial hydrolysis and condensation of silane. Compared to linear polysiloxanes with high chain flexibility, the remaining silanol groups of stereoscopic oligosiloxane nanoclusters can cross-link the polyurea chains into a dense hybrid network and interlink the network to the substrate. Meanwhile, the low-surface-energy RAP can migrate onto the surface because of the self-enriched ability driven by thermodynamic and kinetic driving forces, and its terminated silanol can condense with the above polymer network during coating formation. This self-enriched ability provides the coating surface with superior fouling resistance against proteins and bacteria, while the urea and silane groups provide the coating bottom with strong substrate adhesion. The relationship between the composition of the coating and its mechanical strength, substrate adhesion, and fouling resistance has been investigated. We aimed to propose a self-enrichment strategy to achieve the coexistence of superior antifouling properties and strong substrate adhesion in a coating for potential applications in flexible electronics, biomedical instruments, and marine industries.

■ EXPERIMENTAL SECTION

Materials. Methyltriethoxysilane (MTES), phenyltriethoxysilane (PTS), poly(propylene glycol) bis(2-aminopropyl ether) (D2000, $M_n = 2000 \text{ g mol}^{-1}$), isophorone diisocyanate (IPDI), 3-aminopropyltrimethoxymethylsilane (KH554), 3-aminopropylmethyldiethoxysilane (KH902), poly(ethylene glycol) methyl ether methacrylate (PEGMA, $M_n = 475 \text{ g mol}^{-1}$), and (3-mercaptopropyl) methyldimethoxysilane (KH580) were all procured from Aladdin and used as received. Dichlorodimethylsilane (DCDMS) and fibrinogen ($M_w = 340 \text{ kDa}$) were obtained from Merck Chemicals. Hydrogen chloride solution (HCl, 37 wt %) and deionized water were procured from Sinopharm and used as received. Ethanol, ethyl acetate, and methanol were obtained from Tianjin Fuyu Fine Chemical. Dodecafluoroheptyl methacrylate (DFMA), tetrahydrofuran (THF), and *n*-hexane were procured from Energy Chemical. Azobis(isobutyronitrile) (AIBN) was procured from Aladdin and purified by recrystallizing twice from methanol.

Synthesis of SPU. D2000 (12.00 g) in 30 mL of THF was added dropwise into a solution of IPDI (2.67 g) dissolved in 14 mL of THF. After stirring at 10 °C for 3 h, KH554 (1.96 g) in 6 mL of THF was added dropwise into the above solution and then reacted at an ambient temperature for another 3 h. The solvent was removed from the resulting solution using rotary evaporation.

Preparation of SPU- N_x Coatings. The polysiloxane coatings were prepared by using oligosiloxane nanoclusters and SPU. The obtained polysiloxane coatings were designated as SPU- N_x (x are the weight ratios of MTES and PTS to SPU). Typically, SPU- N_{50} was synthesized using the following steps. MTES (7.10 g), PTS (14.40 g), HCl solution (30.00 μL), ethanol (17.00 mL), and H_2O (2.30 mL) were mixed in a three-neck flask using a magnetic stirrer. The above mixture was reacted at 80 °C for 5 h before being concentrated under

a vacuum to remove volatiles. Subsequently, SPU (21.50 g) and THF (60.00 mL) were added to the above oligosiloxane nanoclusters to form a homogeneous solution, which was followed by adding alkaline catalyst KH902 (2.10 mL) and H_2O (2.10 mL). KH902 is a typical silane, which can participate in the cohydrolysis/condensation with SPU or nanoclusters. Meanwhile, it can act as an alkaline catalyst by an amino group for the further condensation of polysiloxane coatings.⁴¹ After continuous magnetic stirring at 80 °C for 0.5 h, 2.3 g of the prepolymer solution (45 wt %) was cast onto substrates ($10 \times 10 \times 0.5 \text{ cm}$) and then reacted at 50 °C and a 70% relative humidity (RH) for another 3 days to obtain the polysiloxane coatings. The tested substrates included glass slides (silicate glass), ceramic panels (alumina ceramics), steel panels (Q235 steel), Ti panels (pure titanium TA2), and epoxy panels (fiberglass epoxy resin) and were washed with ethanol before being coated. The thickness of the coating was controlled by the relationship between dosage and coating area. The resulting coatings are $\sim 90 \mu\text{m}$ thick. SPU- N_0 , SPU- N_{30} , and SPU- N_{70} were also prepared following the same procedures, with the mass ratio of silane (MTES and PTS) to SPU being 0:100, 30:70, and 70:30, respectively.

Synthesis of the Oligosiloxane Nanoclusters with Different Degrees of Condensation (DOC). An appropriate molar concentration of water is required to obtain oligosiloxane nanoclusters with varying degrees of condensation (DOC). Hence, the reactions were vigorously performed with the same amount of solvent and catalyst, using the following conditions: the molar ratios of silane (MTES and PTS): water are 5:0, 5:3, 5:6, and 5:9. Subsequently, the polysiloxane coatings with different DOCs were prepared using the same preparation method as the SPU- N_x coating.

Synthesis of RAP. First, the telomer was synthesized according to the literature.⁴² KH580 (0.90 g), PEGMA (6.67 g), DFMA (3.34 g), AIBN (0.02 g), and THF (20.00 mL) were added to a Schlenk tube. After three freezing–pumping cycles, the tube was filled with argon. Next, the above solution was reacted for 24 h at 70 °C. The telomer was obtained as a colorless liquid after precipitating in *n*-hexane and then drying under a vacuum at 40 °C. According to the ^1H NMR, the molar mass was calculated to be $\sim 2600 \text{ g mol}^{-1}$. Subsequently, telomer (4.00 g), DCDMS (4.00 g), and THF (20.00 mL) were added dropwise into 80 mL of deionized water in a flask, and the reaction was allowed to run for 12 h at 80 °C under vigorous stirring. Then, a saturated NaHCO_3 solution was added to neutralize the HCl produced during the reaction. The above mixture was subsequently extracted by ethyl acetate and dried with sodium sulfate. Finally, an oily product was obtained after filtration and concentration. The number average molecular weight of RAP is 4600 g mol^{-1} and polydispersity is 1.8.

Preparation of SPU- N_{50} -RAP $_y$ Coatings. Polysiloxane coatings with different RAP contents were synthesized using RAP to replace a part of the SPU. Typically, SPU- N_{50} -RAP $_{5.0}$ was prepared by the following conditions: the weight ratios of silane (MTES and PTS):SPU:RAP become 50:45:5. The antifouling polysiloxane coatings were obtained using the same preparation method as the SPU- N_{50} coating.

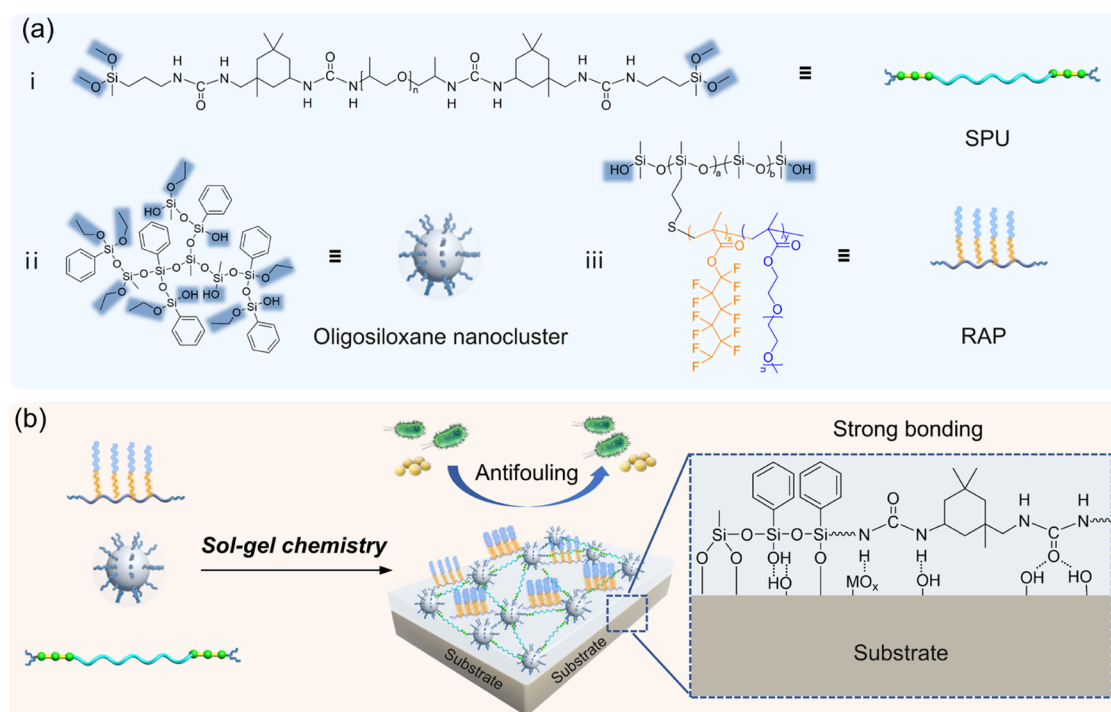
■ CHARACTERIZATION

Nuclear Magnetic Resonance (NMR) Spectroscopy. ^1H NMR spectra and ^{29}Si NMR spectra were obtained on a Bruker AV600 NMR spectrometer using CDCl_3 as the solvent and tetramethylsilane as the interior label.

Transmittance Spectral Measurements. All samples were coated on cleaned glass substrates with a thickness of $\sim 90 \mu\text{m}$. The UV–vis spectra were recorded using an Agilent Model Cary 35500 UV–vis spectrophotometer.

Adhesion Tests. Adhesion tests were performed using a pull-off adhesion tester (PosiTest AT–A Automatic) according to ASTM D4541-09. Five coated substrates ($10 \times 10 \times 0.5 \text{ cm}$) including glass slides (silicate glass), ceramic panels (alumina ceramics), steel panels (Q235 steel), Ti panels (pure

Scheme 1. (a) Chemical Structures of SPU, Oligosiloxane Nanocluster, and RAP, (b) Schematic Description of the Coating Fabrication Process



titanium TA2), and epoxy panels (fiberglass epoxy resin) were used for adhesion tests. The thicknesses of the coatings are $\sim 90 \mu\text{m}$. Before the adhesion test, the coating surface was lightly polished with abrasive paper to improve the adhesion of epoxy-adhesive-coated aluminum studs (20 mm in diameter). The rate of pull was set to 0.2 MPa s^{-1} . The average adhesion strength was calculated based on five different regions of the coatings.

Tensile Test. A tensile test was performed by an Instron 5966 universal mechanical testing machine with a 10 kN load cell at 25°C to evaluate the mechanical properties. The sample was prepared by pouring the prepolymer solution into a poly(tetrafluoroethylene) (PTFE) mold and kept at 50°C and 70% RH for another 3 days to form a polysiloxane film. Then, the film was then peeled away from the PTFE mold and cut into rectangular sections (10 mm in width, 0.4 mm in thickness, and 50 mm in gage length) for further testing. The rectangular sample was stretched with a constant rate of $25 \text{ mm}\cdot\text{min}^{-1}$. Cyclic tensile tests were performed by stretching the samples to 100% strain and then unloading them to the original position with a constant strain rate of $25 \text{ mm}\cdot\text{min}^{-1}$.

Dynamic Mechanical Test. Dynamic mechanical analyses (DMA) were conducted using a NETZSCH DMA 242C in the tensile test mode under a nitrogen atmosphere with a heating rate of $10^\circ\text{C min}^{-1}$ from -80 to 140°C at a constant frequency of 1 Hz. The film samples are 5 mm in width, 0.4 mm in thickness, and 10 mm in gage length.

Surface Wettability Measurements. The static contact angle and dynamic water receding contact angle (θ_R) of the coatings ($\sim 90 \mu\text{m}$ in thickness) were measured using a contact angle goniometer (Biolin, Model KSV NIMA) at 25°C . The droplet volume employed for static contact angles was $3 \mu\text{L}$ of liquid. The θ_R was measured by withdrawing a droplet rest on the coating until it receded.

X-ray Photoelectron Spectroscopy (XPS). XPS measurements were performed to analyze the elemental compositions on the coating surfaces using an Axis Ultra DLD (Kratos Analytical) with a monochromatic Al K α ($h\nu = 1486.6 \text{ eV}$) source at takeoff angles of 90° . The thicknesses of the coatings are $\sim 90 \mu\text{m}$.

Gel Permeation Chromatography. The test was conducted at 35°C with two identical PLgel columns (5 mm, MIXED-C), and THF was chosen as the eluent with a flow rate of $1.0 \text{ mL}\cdot\text{min}^{-1}$. Pretests on polystyrene standards with narrow polydispersity were used to determine the calibration curve.

Dynamic Light Scattering (DLS). The characterization of the average oligosiloxane nanocluster size was performed using a Zetasizer Nano (HORIBA SZ-100Z). Ethanol with a refractive index of 1.45 was used as a solvent.

Quartz Crystal Microbalance with Dissipation (QCM-D). The protein resistance of SPU- N_{50} -RAP $_y$ was investigated using a QCM-D from Q-sense AB (Sweden). In brief, the shift in frequency (Δf) is correlated with the protein adsorption on quartz crystal, whereas the shift in dissipation (ΔD) is integrated with the viscoelastic properties of the adsorbed protein layer. The samples were prepared by spin-casting (2000 rpm, 20 s) an SPU- N_{50} -RAP $_y$ solution in THF (1.0 wt %) on an AT-cut quartz crystal and then cured at 50°C and 70% RH for 3 days. The fibrinogen solution (1 mg mL^{-1}) was delivered to the surface using PBS as a reference at 25°C . The uncoated AT-cut quartz crystal was used as a control.

Bacterial Adhesion Assays. Typical Gram-positive and Gram-negative bacteria, including *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*), were employed for the antibacterial test. A coated Si wafer ($1 \times 1 \text{ cm}$) was immersed in 1 mL of bacterial suspension of *E. coli* or *S. aureus* with a concentration of 10^5 – 10^6 colonies forming unit (CFU)/mL under static conditions for 4 h. Afterward, the coated silicon

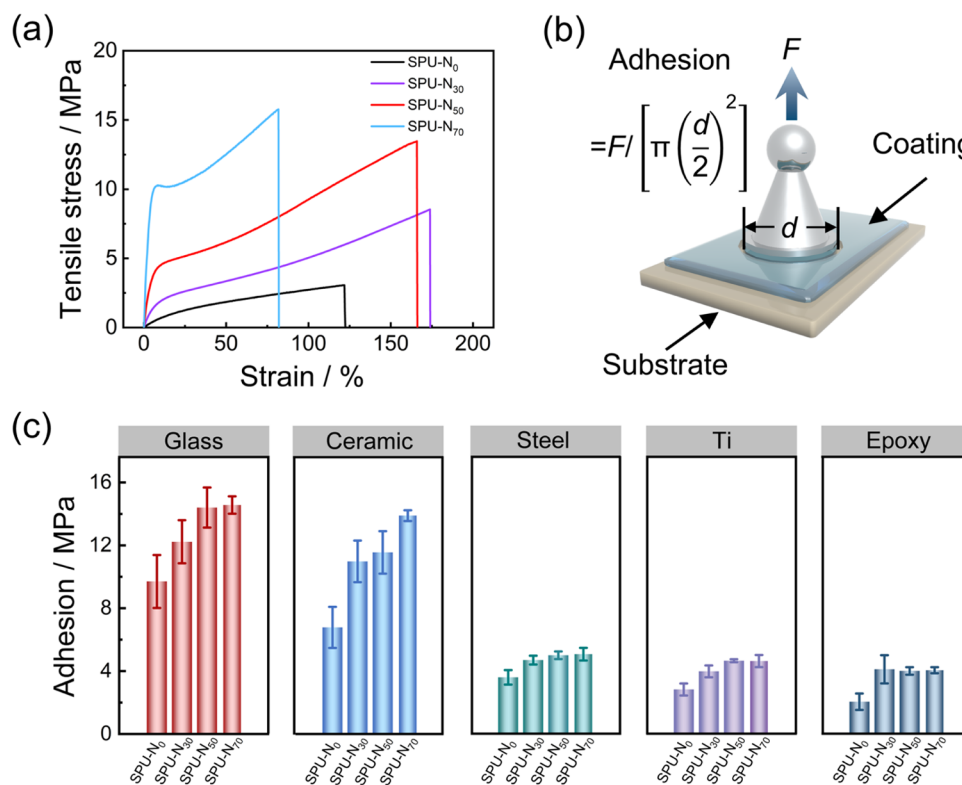


Figure 1. (a) Stress–strain curves of SPU-N_x coatings. (b) The methods employed for the adhesion tests. (c) The adhesion strength of SPU-N_x coatings on various substrates.

wafer was washed gently with PBS to remove the unattached bacteria. The coated silicon wafer was then placed in a tube with 1 mL of PBS, and the strongly adhered bacteria were collected by ultrasonication for 1 min. After that, 100 μ L of the above bacterial suspension was spread onto the beef protein medium and incubated at 37 °C for 24 h. The number of adhered bacteria was calculated by multiplying the number of colonies and the dilution factor. The uncoated Si wafer was used as a control.

Live–Dead Bacterial Assessment. Bacterial log reduction tests were used to evaluate the bactericidal activity of the coatings as described before.^{43,44} A coated 48-well Sterilin plate wafer (0.9 cm²) was immersed in 1 mL of bacterial suspension of *E. coli* or *S. aureus* with a concentration of 10⁴–10⁵ colony forming unit (CFU)/mL under static conditions for 24 h. After ultrasonication for 1 min, 100 μ L of the above bacterial suspension was spread onto the beef protein medium. The number of living bacteria was calculated by multiplying the number of colonies and the dilution factor after incubating at 37 °C for 24 h. The uncoated 48-well Sterilin plate wafer was used as a control.

RESULTS AND DISCUSSION

Scheme 1 depicts the synthetic route for the preparation of polysiloxane coating *via* sol–gel chemistry between SPU, oligosiloxane nanocluster, and RAP. **Figures S1 and S2** present the synthetic route and ¹H NMR of the SPU. Peaks at 0.12 and 3.56 ppm can be assigned to the methyl group and methoxy group on silane. Peaks at 5.89 and 6.28 ppm are attributed to the proton of urea groups. The signal of CH₃ and CH₂ of the D2000 moiety can be observed at 1.11 and 3.47 ppm, respectively. These results indicate the successful synthesis of the SPU. **Figure S3** shows the synthetic route of the

oligosiloxane nanocluster. Furthermore, oligosiloxane nanoclusters with varying DOCs can be readily prepared using the hydrolysis/condensation reaction of PTS and MTES at different water dosages. **Figure S4** shows that the original characteristic signals (T_0) of PTS and MTES in the ²⁹Si NMR spectrum disappeared as the water dosage increased. The terminal units (T_1), linear units (T_2), and dendritic units (T_3) of PTS and MTES appeared in sequence, demonstrating that higher water dosage favors higher condensation degrees of oligosiloxane nanoclusters. The DOC of the oligosiloxane nanoclusters was calculated using the following equation

$$\text{DOC} = \frac{T_1 + 2T_2 + 3T_3}{3 \times (T_1 + T_2 + T_3)} \times 100\%$$

The calculated DOC was 0, 54, 74, and 90% for silane and water molar ratios of 5: 0, 5: 3, 5:6, and 5:9, respectively, illustrating the different DOCs of successfully synthesized oligosiloxane nanoclusters. Note that the nanoclusters are used as a binder rather than fillers and they can condense with other hydroxyl or alkoxy groups to generate water or alcohol that will volatilize in the air, so the volume content used to denote the weight of the filler per unit volume of coating cannot be obtained.^{45,46} The as-prepared oligosiloxane nanoclusters were subsequently added into SPU as cross-linking points *via* cohydrolysis/condensation reaction to generate a polysiloxane coating. A tensile test was used to systematically investigate the influence of the DOC of oligosiloxane nanoclusters on the coating. **Figures S5 and S6** show the typical tensile stress–strain curves of SPU-N₇₀ with different DOCs. As the DOC increases, the coating exhibits higher tensile strength but lower elongation at the break because of the increased cross-linking density. Despite having the highest tensile strength (15.90

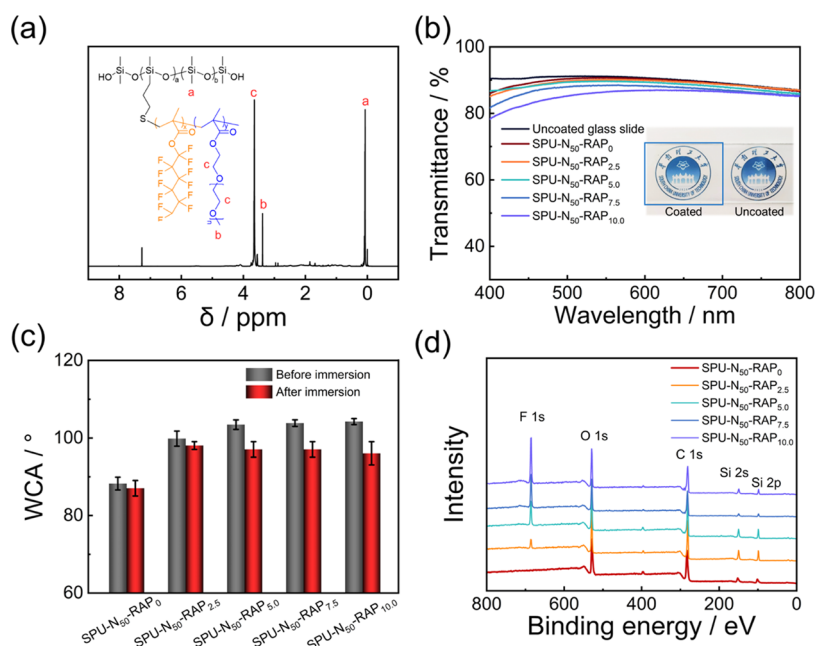


Figure 2. (a) ¹H NMR spectrum of RAP. (b) Optical transmission measurements of the uncoated and coated glass slides with a thickness of ~ 90 μm . (c) Static water contact angles (WCAs) of SPU-N₅₀-RAP_y coatings before and after immersion in water for 24 h. (d) XPS scan spectra of SPU-N₅₀-RAP_y coatings.

MPa), the SPU-N₇₀ with 90% of DOC exhibited hard elastic properties. Hence, the SPU-N₇₀ with 74% of DOC, which exhibited an ~ 15.35 MPa tensile stress and $\sim 82\%$ elongation, was selected for further investigation.

The mechanical properties of polysiloxane coatings with various oligosiloxane nanoclusters/SPU ratios were measured by tensile and adhesion tests. The nanocluster with a DOC of 74% and an average size of ~ 1.6 nm was used (Figure S7). Figures 1a and S8 depict that the tensile strengths and elastic modulus of the polysiloxane coatings increase significantly with the increase in the oligosiloxane nanocluster content. Notably, the elongation at break first increases and then decreases. A probable reason is that the proper content of oligosiloxane nanoclusters increased the cross-linking density, thereby enhancing the strong H-bonding interaction. To prove this, the cross-linking density of the SPU-N_x coating was calculated via the following equation based on the statistical theory of rubber elasticity⁴⁷

$$\rho = \frac{E}{3RT}$$

where E is the storage modulus of elastomers in the rubbery plateau region at the temperature T and R is the gas constant. Based on the values of storage modulus (Figure S9), the corresponding ρ of SPU-N₀, SPU-N₃₀, SPU-N₅₀, and SPU-N₇₀ was 540, 924, 1491, and 8873 mol m⁻³, respectively, indicating that the higher oligosiloxane nanocluster content, the higher the cross-linking density. However, the high oligosiloxane nanocluster content adversely affects the molecular growth and increases the hard segment content of the coating, compromising its elongation at break.^{48,49}

A pull-off adhesion tester was used to measure the adhesion strength of the polysiloxane coatings coated on various substrates (Figure 1b). Figure 1c shows that SPU-N₀ has the lowest adhesion strength on epoxy (2.06 ± 0.52 MPa), Ti (2.83 ± 0.38 MPa), steel (3.60 ± 0.46 MPa), ceramic ($6.78 \pm$

1.30 MPa), and glass (9.70 ± 1.68 MPa). Notably, when oligosiloxane nanoclusters were introduced to the SPU, the adhesion increased such that the adhesion strengths of SPU-N₅₀ on epoxy, Ti, steel, ceramic, and glass were 4.01 ± 0.24 , 4.66 ± 0.10 , 5.01 ± 0.24 , 11.55 ± 1.35 , and 14.40 ± 1.27 MPa, respectively. The comparatively higher adhesion on glass and ceramic substrate could be attributed to plenty of reactive silanol groups hydrolyzed from alkoxy groups on oligosiloxane nanoclusters, which can cross-link the SPU into a tough polymer network by sol–gel condensation and interlink the polymer network to the substrate by condensation with the hydroxyl groups of the substrate surfaces during the curing process. In addition, the urea groups also form strong hydrogen bonding between polysiloxane coating and polar substrates (e.g., glass and ceramic), further improving the adhesion on substrates. Regarding the epoxy, Ti, and steel substrates, strong substrate adhesion is mainly attributed to strong hydrogen bonding with polar groups of these surfaces.⁵⁰ In the following experiment, we used SPU-N₅₀ as the basis to endow it with antifouling ability due to its excellent mechanical properties and strong adhesion on substrates.

The bis-silanol-terminated siloxane with telomer side chains was designed as a RAP, where the above surface-enriched amphiphilic telomer side chain is for fouling-resistant function, and the soft siloxane main chain is for improving surface enrichment without compromising on the mechanical properties of the coating. The synthetic routes of telomer and RAP are shown in Figures S10 and S11, and the successful synthesis of RAP was confirmed via ¹H NMR (Figures 2a and S12). The transmittance spectra of polysiloxane coatings coated on glass slides are shown in Figure 2b. All of the polysiloxane coatings with a thickness of ~ 90 μm show an average transmittance of $>85\%$ at a 500 nm wavelength, which was attributed to the size of nanoclusters and RAP phase domain being smaller than the visible light wavelength. The inset in Figure 2b illustrates a colorful picture covered by the sample SPU-N₅₀-RAP_{5.0} with no reduction in optical clarity, indicating superior trans-

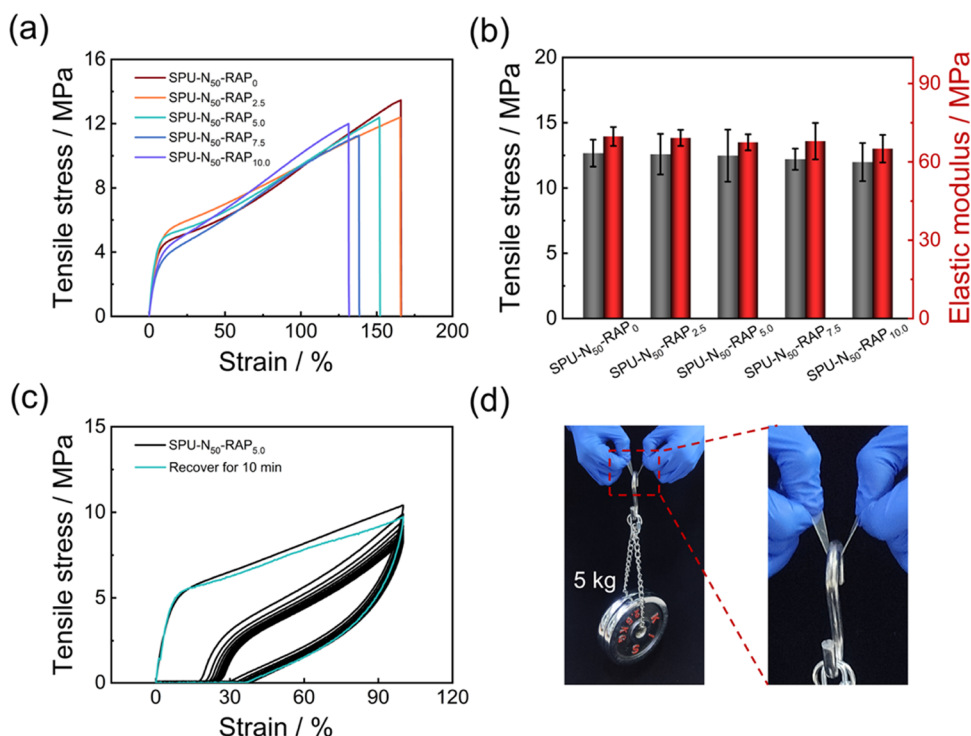


Figure 3. (a) Stress–strain curves of SPU-N₅₀-RAP_y and (b) the corresponding tensile stress and elastic modulus. (c) Cyclic stretch curves of SPU-N₅₀-RAP_{5.0}. (d) The images depict the load-bearing capacity of SPU-N₅₀-RAP_{5.0}.

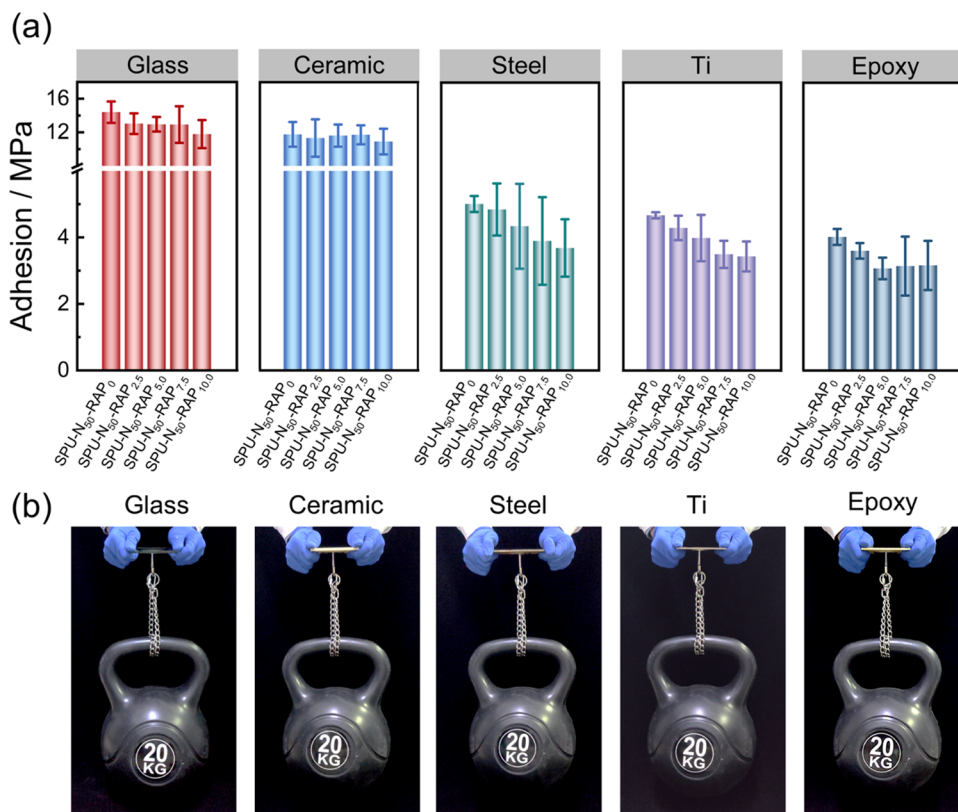


Figure 4. (a) Adhesion strength of SPU-N₅₀-RAP_y coatings on various substrates. (b) The images show that the SPU-N₅₀-RAP_{5.0} cast on substrates can withstand a full force of 20 kg.

parency. Note that the transparency decreased as the amount of reactive amphiphilic polymer increased; this is because the siloxane main chain and fluorocarbon side chains of the RAP

have poor compatibility with the basic coating, and thus high RAP content easily leads to phase separation.

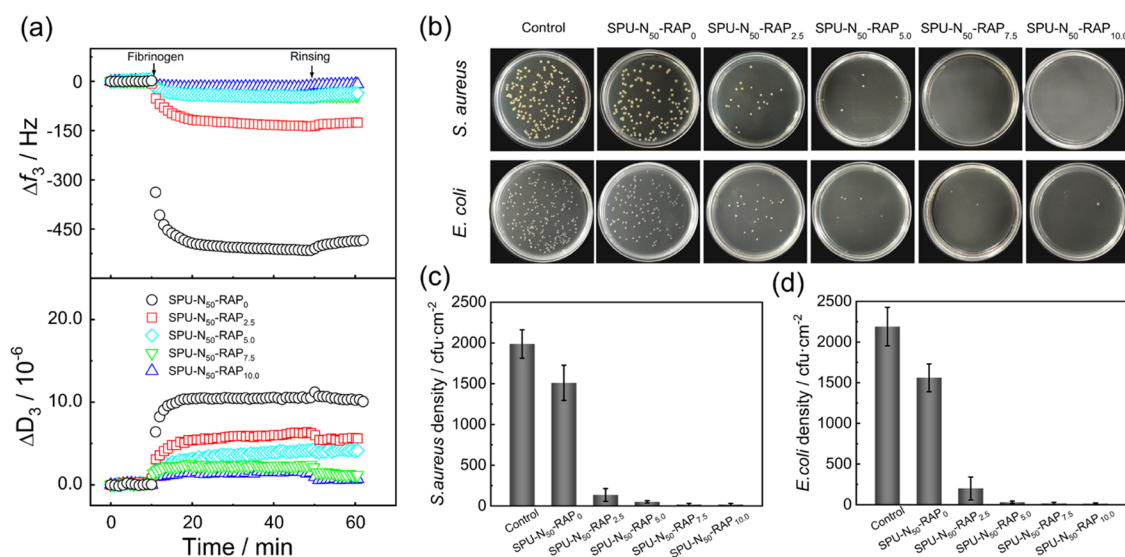


Figure 5. (a) QCM-D curves of SPU-N₅₀-RAP_y coatings. (b) The photographs of the bacterial colonies on the surfaces of the polysiloxane coating with different contents of RAP. The corresponding density of the attached bacterial numbers of (c) *S. aureus* and (d) *E. coli*.

To further confirm the successful loading RAP in SPU-N₅₀, the static water contact angle (WCA) of the polysiloxane coating before and after immersion was investigated. As shown in Figure 2c, the static WCAs of coatings increased as the RAP content increased but decreased after immersion in deionized water for 24 h. Note that there is not much difference in the static WCA before and after water immersion due to the low content of PEG. We have conducted the dynamic WCAs of the polysiloxane coatings before and after immersion (Figure S13). It is well known that the dynamic water advancing contact angles are related to the hydrophobicity of the surface, whereas the dynamic water receding contact angles (θ_R) are related to the hydrophilicity of the surface.⁵¹ The θ_R of coatings decreased sharply after immersion in deionized water. The large difference in the θ_R before and after immersion in water clearly indicated that the PEG segments had rearranged to the surface. Furthermore, XPS spectra are conducted to characterize the surface composition of polysiloxane coatings (Figure 2d). The peak at 685.7 eV is attributed to F (1s) in the fluorocarbon segment of RAP. For SPU-N₅₀-RAP₀, no F (1s) can be detected because it does not contain any fluorocarbon, and the peak intensity of F (1s) increases with the RAP content. More importantly, all of the polysiloxane coatings except SPU-N₅₀-RAP₀ exhibit a higher surface fluorine atomic composition than those in bulk (Table S1). This surface enrichment of fluorine indicates that the RAP can be migrated to the surface using the low surface energy of fluorocarbon as a driving force.

Enrichment of the grafted RAP on the coating surface can achieve superior antifouling ability without compromising the mechanical properties of the coating. Figure 3a depicts that the elongation at break decreases as the content of the RAP increases because some of the SPU segments are replaced by RAP with multiple side chains, resulting in less dynamic hydrogen-bonding interaction of the system. The tensile strengths and elastic modulus of the polysiloxane coatings change negligibly when the soft bis-silanol-terminated siloxane main chain of RAP is utilized (Figure 3b). Furthermore, SPU-N₅₀-RAP_{5.0} was chosen as the typical sample for the cyclic tensile test, as shown in Figure 3c. After 10 cycles of tension, SPU-N₅₀-RAP_{5.0} remains intact and almost returns to its

original state with a 10 min recovery (recovery rate $\sim 88\%$), showing better resilience and fatigue resistance. Moreover, the SPU-N₅₀-RAP_{5.0} (~ 10 mm width, ~ 0.2 mm thickness) can withstand a 5 kg dumbbell for more than 10 min without breaking (Figure 3d).

Furthermore, as RAP content increases, the adhesion strength on various substrates is only slightly affected (Figure 4a). This is because the low-surface-energy RAP can migrate onto the surface via the self-enriched ability, and the urea and silane groups provide the coating bottom with strong substrate adhesion. However, the adhesion strength decreases with increasing RAP content, especially on steel, Ti, and epoxy, which can be attributed to less hydrogen bonding between coatings and substrates resulting from changing SPU with urea groups to RAP with amphiphilic side chains. Anyhow, the SPU-N₅₀-RAP_{10.0} retained its high adhesion to glass (~ 11.79 MPa), ceramic (~ 10.90 MPa), steel (~ 3.68 MPa), Ti (~ 3.43 MPa), and epoxy (~ 3.16 MPa). Such adhesion strength can be readily used in practical applications without a tie-coat. Moreover, Figure 4b shows that a steel hook bonded to various substrates with SPU-N₅₀-RAP_{5.0} can withstand a full force of 20 kg.

Since the formation of biofouling usually begins with protein adsorption and many fouling organisms secrete protein-based bioadhesives to stick to surfaces,⁴² we investigated the protein resistance of the polysiloxane coatings using QCM-D. Figure 5a shows the time dependences of frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on SPU-N₅₀-RAP_y coatings. For SPU-N₅₀-RAP₀, it has an obvious decrease in Δf and an increase in ΔD after rinsing with PBS, indicating the considerable adsorption of fibrinogen on the SPU-N₅₀-RAP₀ surface. SPU-N₅₀-RAP_{2.5} shows a large lower decrease of Δf and an increase in ΔD , exhibiting limited protein resistance capacity. In contrast, SPU-N₅₀-RAP_{5.0}, SPU-N₅₀-RAP_{7.5}, and SPU-N₅₀-RAP_{10.0} exhibit superior protein resistance reflected in the almost negligible changes in Δf and ΔD .

We also conducted antibacterial assays using the disk diffusion method to verify the antibacterial properties of the polysiloxane coatings. As shown in Figure 5b, a large amount of alive *S. aureus* and *E. coli* colonies are observed in the

control plates (*S. aureus* ~ 1980 cfu·cm $^{-2}$, *E. coli* ~ 2190 cfu·cm $^{-2}$) and SPU-N₅₀-RAP₀ plates (*S. aureus* ~ 1510 cfu·cm $^{-2}$, *E. coli* ~ 1560 cfu·cm $^{-2}$), indicating that it is very susceptible to bacterial adhesion. The number of colonies of the coated sample decreased significantly as the RAP content increased, and almost no bacteria (*S. aureus* <20 cfu·cm $^{-2}$, *E. coli* <20 cfu·cm $^{-2}$) were observed on the plates of SPU-N₅₀-RAP_{7.5} and SPU-N₅₀-RAP_{10.0}, implying that the self-enriched amphiphilic telomer side chain was conducive to enhancing the antibacterial property of the surface (Figure S5c,d). Moreover, bacterial log reduction tests were used to evaluate the bactericidal activity of the coatings. As shown in Figure S14, the number of colonies of *S. aureus* and *E. coli* of SPU-N₅₀-RAP_{10.0} is almost the same as the control sample, indicating that the polysiloxane coatings exhibited superior antifouling ability via antibacterial adhesion instead of killing bacteria. With this unique antibacterial property, our coating may provide a good prospect for preventing bacterial adhesion in the maritime industry and biomedical engineering.

As discussed above, most silicone coatings do not usually exhibit both superior antifouling ability and strong substrate adhesion, as shown in Table S2. In this study, the low-surface-energy RAP can migrate onto the surface via the self-enriched ability, and the urea and silane groups provide the coating bottom with strong substrate adhesion. Such a coating exhibits superior antifouling ability and strong substrate adhesion. Therefore, the self-enrichment method provides a promising strategy to solve the contradiction of superior antifouling ability and strong substrate adhesion in a silicone coating.

CONCLUSIONS

We prepared a transparent polysiloxane coating with superior antifouling performance and strong substrate adhesion via the sol–gel chemistry of SPU, oligosiloxane nanocluster, and RAP. We found that low-surface-energy amphiphilic segments can self-enrich onto the surface, whereas strong hydrogen bonds and silane can interlink the coating bottom to substrates during film formation. The resulting coating has superior fouling resistance against proteins and bacteria, high transparency, tensile strength, and strong substrate adhesion. This research paves an effective way to develop high-performance polysiloxane antifouling coatings. Such a coating is expected to find application in flexible electronics, stretchable optical devices, marine industries, and other fields.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.3c00199>.

Synthetic route and ^1H NMR spectrum of the SPU; synthetic route of the oligosiloxane nanocluster; ^{29}Si NMR spectra of the oligosiloxane nanoclusters with silane to water molar ratios of 5:0, 5:3, 5:6, and 5:9; stress–strain curves and tensile stress of SPU-N₇₀ coatings with different degrees of condensation; size distribution of oligosiloxane nanoclusters in the DLS analysis; tensile stress and elastic modulus of SPU-N_x coatings; dynamic mechanical curve of SPU-N_x coatings; synthetic route of the telomer and RAP; ^1H NMR spectrum of the telomer; θ_{R} of SPU-N₅₀-RAP_y coatings before and after immersion in water for 24 h; live–dead assessment of the coatings; the elemental mass content

of F on the coatings; and comparisons of substrate adhesion and antifouling ability of silicone coatings (PDF)

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Notes

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